

What is probability?

Probability is just a number between 0 and 1 that says how likely something is. 0 means it will never happen. 1 means it will definitely happen. 0.5 means a fair coin flip.

Analogy: Think of probability as the percentage of times something happens if you repeat an experiment a million times. If it rains 30 days out of 100, $P(\text{rain}) = 0.30$.

The basic rules:

$0 \leq P(A) \leq 1 \leftarrow$ prob always between 0 and 1
 $P(A) + P(\text{not } A) = 1 \leftarrow$ something either happens or it doesn't

$P(A \text{ or } B) = P(A) + P(B) \leftarrow$ addition rule

$P(A \text{ and } B) = P(A) \times P(B|A) \leftarrow$ multiplication rule

In ML: A classifier outputs $P(\text{cat} | \text{image}) = 0.87$, meaning "87% confident this is a cat." That IS probability.

Conditional probability

It measures the likelihood of an event occurring given that another related event has already taken place. It is denoted as $P(A|B)$, which reads as "the probability of A given B"

The Formula

To calculate conditional probability, you divide the probability of both events occurring together (their intersection) by the probability of the condition (the given event).

Conditional Probability Formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B

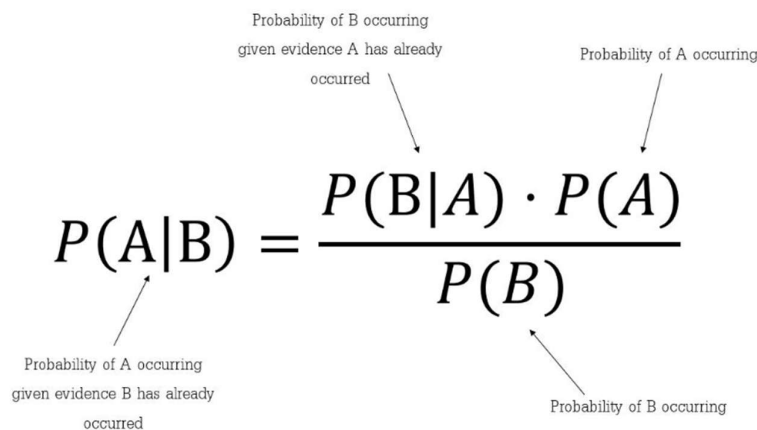
Probability of A and B

Probability of B

P(A and B) – Joint Probability

Bayes' Theorem

The most important formula in ML. It lets you update your belief when new evidence arrives.



The diagram shows the Bayes' Theorem formula with arrows pointing from descriptive text to the corresponding parts of the formula:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

- An arrow points from "Probability of B occurring given evidence A has already occurred" to $P(B|A)$.
- An arrow points from "Probability of A occurring" to $P(A)$.
- An arrow points from "Probability of A occurring given evidence B has already occurred" to $P(A|B)$.
- An arrow points from "Probability of B occurring" to $P(B)$.

$$\text{Posterior} = (\text{Likelihood} \times \text{Prior}) / \text{Evidence}$$

Where:

Prior = what you believed before seeing data

Likelihood = how probable this data is if A is true

Evidence = total probability of seeing this data

Posterior = updated belief after seeing the data

Statistics and Its Role in Machine Learning

Statistics is the study of collecting, analyzing, interpreting, and summarizing data. It helps us understand patterns, make decisions, and draw conclusions from information.

In machine learning, statistics is extremely important because ML models learn patterns from data. Without statistics, it is difficult to understand the quality, distribution, and behavior of the data being used.

Types of Statistics

Statistics is mainly divided into two categories:

Descriptive Statistics

Descriptive statistics summarizes and describes the data you already have.

It answers questions like:

- What is the average value?
- How spread out is the data?
- What is the middle value?

Common measures include:

- Mean
- Median
- Variance
- Standard deviation

Example

Suppose student marks are:

70,75,80,85,90

The mean tells us the average score, while standard deviation tells us how consistent the scores are.

Analogy

Descriptive statistics is like a report card summary. Instead of looking at every assignment individually, you look at averages and overall performance.

Inferential Statistics

Inferential statistics uses a small sample of data to make predictions or conclusions about a larger population.

Instead of studying every person or object, we analyze a sample and estimate the overall trend.

Example

A company surveys 1,000 people to predict how millions of customers may react to a product.

Common Concepts

- Probability
- Hypothesis testing
- Confidence intervals
- Sampling

Analogy

If you taste one spoon of soup to decide whether the whole pot needs salt, you are using inferential statistics.

Statistics in Machine Learning

Machine learning heavily relies on statistics because models learn from numerical patterns in data.

Statistics helps in:

- Understanding data distribution
- Detecting outliers
- Feature scaling
- Measuring model performance
- Estimating probabilities

Many ML algorithms are built directly on statistical concepts.

Important Statistical Concepts in ML

Mean

The mean is the average value of a dataset.

Formula

$$\mu = \frac{1}{n} \sum x_i$$

Where:

- μ = mean
- x_i = each value in the dataset
- n = total number of values
- $\sum$ = summation symbol

Use in ML

The mean is commonly used in feature scaling and normalization.

Variance

Variance measures how far data points are spread from the mean.

Formula

$$\sigma^2 = \frac{1}{n} \sum (x_i - \mu)^2$$

Where:

- σ^2 = variance
- x_i = each data value
- μ = mean
- n = total number of values

Use in ML

Variance helps detect whether features vary enough to be useful for training.

Standard Deviation

Standard deviation is the square root of variance.

Formula

$$\sigma = \sqrt{\sigma^2}$$

Where:

- σ = standard deviation
- σ^2 = variance

Use in ML

Used in standardization:

$$z = \frac{x - \mu}{\sigma}$$

Where:

- z = standardized value (z-score)
- x = original value
- μ = mean
- σ = standard deviation

This transformation makes features easier for models to learn.

Analogy for Statistics in ML

Imagine training a basketball player.

- The **mean** tells you the player's average score.
- The **standard deviation** tells you how consistent the player is.
- Inferential statistics helps predict future performance based on past games.

Machine learning models use similar statistical information to detect patterns and make predictions.

Example: Statistics Used in ML Code

A common statistical operation in ML is feature standardization.

```
from sklearn.preprocessing import StandardScaler
import numpy as np
```

```
# Dataset
```

```
X = np.array([[10], [20], [30], [40], [50]])
```

```
# Create scaler
```

```
scaler = StandardScaler()
```

```
# Standardize data
```

```
X_scaled = scaler.fit_transform(X)
```

```
print(X_scaled)
```

What Happens Here

- The scaler calculates the **mean** and **standard deviation** of the dataset.
- Each value is transformed using:

$$z = \frac{x - \mu}{\sigma}$$

- The resulting data has:
 - Mean = 0
 - Standard deviation = 1

This improves the performance of many ML algorithms such as Logistic Regression, SVMs, and Neural Networks.

2. Median

The **median** is the middle value when data is arranged in order.

Explanation

- If the dataset has an odd number of values → choose the middle value.
- If it has an even number of values → average the two middle values.

Example

Dataset: 2, 4, 7, 9, 12

Median = 7

Important Property

The median is **robust to outliers**, meaning extremely large or small values do not affect it much.

Example with Outlier

Dataset: 2, 4, 7, 9, 100

- Mean = 24.4 (heavily affected)
- Median = 7 (still stable)

Key Idea: Median is often better than mean when data contains outliers.

